PPL FINAL

©Copyright Buğra Akbiz

① Consider the following C declaration, compiled on a 32-bit Pentium machine:

```
struct E
int n;
char c;
} A[10][10];
```

If the address of A[0][0] is 1000(decimal), what is address of A[3][7]?

Answer: $1000 + (3 \times 10 \times 8) + (7 \times 8) = \boxed{1296}$

②
```
int sub(int *i){
  *i += 2;
return(*i) +3; }
void main(){
i=1, j=1, s1, s2;
s1 = sub(&i) + (i+1);
s2 = (j*5) + sub(&j)
```

a) Left to right s1=10 s2=11

b) Right to left s1=8 s2=21

③ Consider where STR is token

A → A Δ B | B

B → C □ B | C

C → (◊ D | D

D → STR ∇ D | STR

Δ: Left

□: right

◊: left

∇: right

Associality of operators?

Scanned by CamScanner

© copyright Bilgin Aktemiz

(4)

```
Procedure main(){
  string x;
Procedure print_message(){
  print(x); 3
procedure sub1(){
  string x = "sub1_message"
  sub2(print_message); 3
Procedure sub2(sx: procedure){
string x = "sub2 message";
 sx(); 3
x = "main message";
 sub1()
```

a) shallow binding

Ans: sub2 message

b) Deep binding

Ans: Main message

Static Scope?
Ad-hoc?
Deep bindingin cevabi dogru mu?

(5)

```
begin
integer n;
procedure p(k:integer);
 begin
n := n+1;
k := k+4;
print(n);
 end;
```

a) Call by value: 1 1

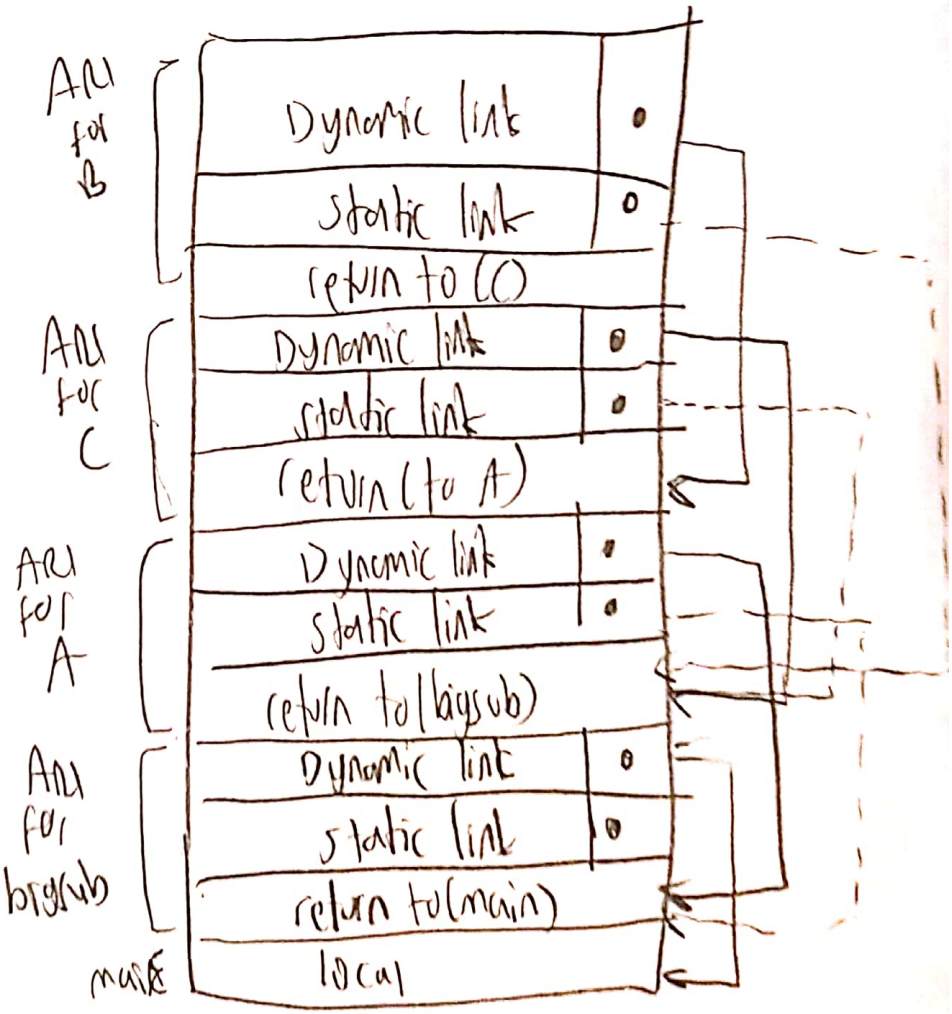
b) Call by reference: 5 5

c) Call by value-result: 1 4

Scanned by CamScanner

(6)

```
procedure Main is
procedure Bigsub is
Procedure A is
Procedure B is
 begin --- of B
  ---  <-------- 1
  end; -- of B
 procedure C is
 begin; -- of C
  --
  B;
 --
 end; -- of C
 begin --of A
  --
  C
 --
 end; -- of A
begin -- of Bigsub
 --
  A;
 --
end; -- of bigsub
```



Answer:

call sequence

$main calls bigsub
bigsub calls A
A calls C
C calls B

Dynamic link → call sequence direction.

Scanned by CamScanner